Inspector-Triggered Ticket-Purchase Fraud Detection

A Hidden Markov Model pipeline over 20 GB of mobile-ticketing logs

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20 GB	6.4 M	98,241	9	100 / 100	88,328
raw input	activation events	HMM-eligible riders	latent states (BIC)	rules ∩ HMM overlap	HMM-only candidate pool

Research Questions

01 DESCRIPTIVE
What does the activation-to-scan gap distribution look like, and is there a measurable sub-population below the fare-evasion cutoff?

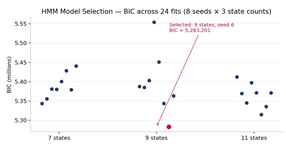
02 MODELLING
Can an HMM fit to per-rider symbol sequences identify a latent fraud-shaped state that rule heuristics miss, and how stable is that structure across model-selection choices?

03 VALIDATION
Where do the two layers (rules and HMM) agree, and what unique population does each identify?

Data Flow — Raw CSVs to Shortlist

- **01 INGEST** — 4 CSVs · 20 GB · 6.4 M events
- **02 SYMBOLIZE** — 7-symbol vocabulary per event
- **03 FIT HMM** — 3 states × 8 seeds · BIC pick
- **04 SCORE** — Posterior · rule · gaming · burst
- **05 SHORTLIST** — R ∩ H · H \ R supplementary

Model Selection — BIC Across 24 Fits

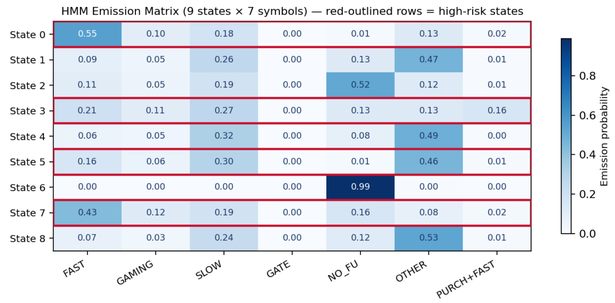


9 states, seed 6 · 31,000 BIC points clear of runner-up.

Symbol Vocabulary (First-Match-Wins)

FAST	handheld scan ≤ 15 s
GAMING	scan 16 – 30 s (dodge band)
SLOW	scan 30 – 300 s
GATE	gate scan ≤ 120 s
NO-FU	no handheld or gate scan
OTHER	fall-through catch-all
PURCHASE→FAST	buy ≤ 60 s before activation

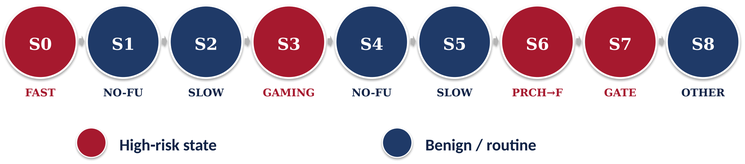
Fitted HMM · 9 States × 7 Symbols



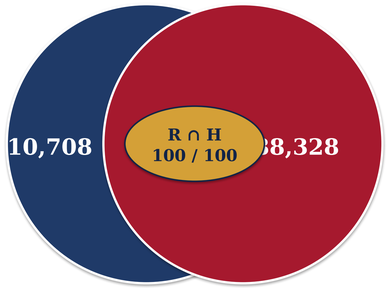
	FAST	GAMING	SLOW	GATE	NO_FU	OTHER	PURCH+FAST
State 0	0.55	0.10	0.18	0.00	0.01	0.13	0.02
State 1	0.09	0.05	0.26	0.00	0.13	0.47	0.01
State 2	0.11	0.05	0.19	0.00	0.52	0.12	0.01
State 3	0.21	0.11	0.27	0.00	0.13	0.13	0.16
State 4	0.06	0.05	0.32	0.00	0.08	0.49	0.00
State 5	0.16	0.06	0.30	0.00	0.01	0.46	0.01
State 6	0.00	0.00	0.00	0.00	0.99	0.00	0.00
State 7	0.43	0.12	0.19	0.00	0.16	0.08	0.02
State 8	0.07	0.03	0.24	0.00	0.12	0.53	0.01

High-risk states (0, 3, 6, 7) concentrate on FAST / GAMING / PURCHASE→FAST.

State Topology — 4 High-Risk / 5 Benign



Rule ∩ HMM — Two Shortlists

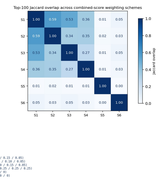


R (rule pool) and H (HMM-only pool) are disjoint by construction; the 100/100 overlap is on the combined top-100 shortlist — evidence the HMM confirms the rules and also surfaces 88K new candidates with zero rule infractions.

Key Findings

1.0000	top posterior dominance (HMM-only shortlist)
0.9999	median posterior dominance (top 100 HMM-only)
74 / 100	shortlist retention under HMM-heavy weighting
Wed · Providence	strongest temporal / geographic concentration

Weight Sensitivity (6 Schemes)



Conclusions & Recommendations

Rules catch the obvious cases; the HMM scores every eligible rider on how often their sequence occupies high-risk latent states — a defensible, model-based extension.

- ↳ **Dispatch inspectors to Providence/Stoughton Wed evenings.**
- ↳ **Retrain the HMM quarterly on a rolling 4-month window.**
- ↳ **Extend activation feed with station_id / line_id for geo features.**

ACKNOWLEDGEMENTS

Charlie Ko (Masabi) and LF Arsenault (Gemsen) for sponsor guidance, data access, and review comments that shaped the HMM-seed sweep, weight-sensitivity analysis, and the symbol-vocabulary design.

REPRODUCIBILITY

Full methodology → UC2_v2_Report.docx
Exact counts, memory, runtimes → RUN_RESULTS.md